PTUT — FINAL YEAR PROJECT

Weekly Progress Report · Week 4 of 12

TaxWise Freelance

Automated Tax Deductions & Invoicing Platform

Database Modeling & UI Wireframes

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SECTION
A

PROJECT ID
PTUT-PRJ-042

DOMAIN
FinTech, Personal Wealth & Micro-Finance

REPORTING PERIOD
4 – 10 November 2026

SUBMITTED
11 November 2026

Source code: github.com/SohaibRamzan01/taxwise-freelance

1. Summary

This week I designed the **database**, the **API** that the app's screens use to talk to the server, and **wireframes** (simple sketches) of the main screens.

9
database tables

30+
REST API endpoints

6
screen wireframes

Task from the plan	Status
Database ERD with index definitions	Done — Figure 1 and section 3
Complete REST API endpoint contracts	Done — section 4
UI wireframes for 4–6 main screens	Done — 6 screens, section 5

2. Entity-Relationship Diagram (ERD)

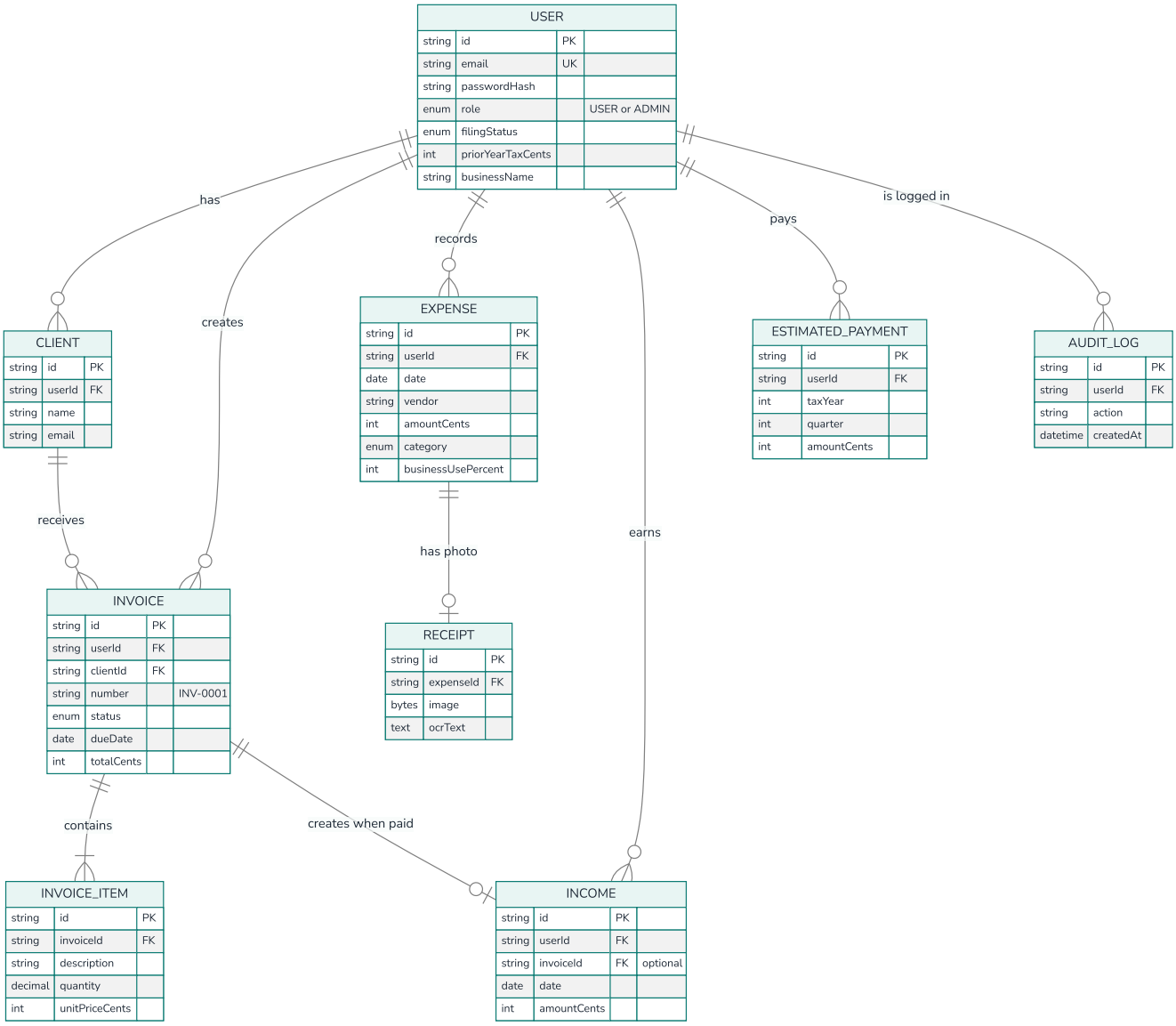


Figure 1 — Entity-Relationship Diagram

Simple rules in the design:

- **Every table has a `userId`.** This keeps each user's data separate (multi-tenant).
- **Money is stored in cents** (whole numbers), e.g. $12.50 is saved as 1250. This avoids rounding errors.
- **Invoice numbers are unique per user** (INV-0001, INV-0002, ...).
- **One paid invoice = one income row.** The database blocks a second one.

3. Indexes

An index makes searching a table fast, like the index at the back of a book. I only added indexes for searches the app really does.

Table	Index	Why it is needed
User	`email` (unique)	Find a user quickly at login; stop duplicate accounts
Expense	`userId, date`	Show a user's expenses for a date range
Expense	`userId, category`	Deductions-by-category chart and filters
Invoice	`userId, number` (unique)	Unique invoice numbers per user
Invoice	`userId, status`	List unpaid / overdue invoices
Invoice	`userId, dueDate`	Upcoming due dates on the dashboard
Income	`userId, date`	Income totals for the year and per month
Income	`invoiceId` (unique)	Makes sure a paid invoice adds income only once
EstimatedPayment	`userId, taxYear, quarter`	Payments made in each quarter
AuditLog	`userId, createdAt`	Show recent activity in the admin portal

4. REST API endpoints

All endpoints use JSON. Every endpoint except sign-up, login and health needs the user to be logged in. Errors always look the same: `{ "error": { "code", "message" } }`.

Area	Method & path	What it does
Auth	`POST /api/auth/signup`	Create an account
Auth	`POST /api/auth/login` · `POST /api/auth/logout`	Log in (sets secure cookie) / log out
Auth	`GET /api/auth/me`	Who is logged in
Settings	`GET` · `PATCH /api/settings`	Tax profile and business details
Expenses	`GET` · `POST /api/expenses`	List (search, filter, sort, pages) / add
Expenses	`GET` · `PATCH` · `DELETE /api/expenses/:id`	View / edit / delete one expense
Receipts	`POST /api/receipts/scan`	Upload a receipt photo; returns vendor, date, total, category
Receipts	`GET /api/receipts/:id`	View the saved receipt photo

Area	Method & path	What it does
Income	`GET` · `POST /api/income`, `PATCH` · `DELETE /api/income/:id`	Manage income
Clients	`GET` · `POST /api/clients`, `PATCH` · `DELETE /api/clients/:id`	Manage clients
Invoices	`GET` · `POST /api/invoices`, `GET` · `PATCH` · `DELETE /api/invoices/:id`	Manage invoices; change status (e.g. mark paid)
Invoices	`GET /api/invoices/:id/pdf`	Download the invoice as PDF
Tax	`GET /api/tax/estimate?year=2026`	Full tax estimate with quarterly schedule
Tax	`GET` · `POST /api/estimated-payments`, `DELETE /api/estimated-payments/:id`	Log tax payments made
Reports	`GET /api/reports/tax-summary?year=2026`	Download the tax-ready summary PDF
Dashboard	`GET /api/dashboard`	Totals, chart data, unpaid invoices, deadlines
Admin	`GET /api/admin/users`, `PATCH /api/admin/users/:id`	List users, change role, activate/deactivate
Admin	`GET /api/admin/stats` · `GET /api/admin/audit-log`	Platform charts and activity log
Health	`GET /api/health`	Is the app and database working?

Example — scan a receipt

Request: `POST /api/receipts/scan` with the photo as form data. Response: `{ "vendor": "Dell", "date": "2026-10-03", "totalCents": 89900, "suggestion": { "category": "COMPUTER_HARDWARE", "confidence": 0.9 } }`

5. Wireframes

1 · Login / Sign up

Logo + "Track deductions, send invoices, plan quarterly tax"

Email

Password

Log in Don't have an account? Sign up

2 · Dashboard

Income YTD	Expenses	Deductions	Tax owed

Chart: income vs expenses by month

Chart: deductions by category

Unpaid invoices

Next tax due: 15 Jan 2027

3 · Expenses + Scan receipt

Scan receipt **+ Add expense** Search · Category ▾ · Date ▾

Table: date · vendor · category · amount · business % · deductible · receipt

Scan dialog: photo preview → vendor / date / total filled in → category "Hardware (auto)" → **Save**

4 · Invoice editor

Client ▾ | Issue date | Due date

Line items: description · qty · price · amount [+ add line]

Subtotal · discount · tax · **Total**

Save draft **Mark sent** **Mark paid** **Download PDF**

5 · Tax estimator

Year ▾ · Filing status ▾ · Safe harbour ☐

Breakdown: profit → SE tax → deductions → taxable income → income tax → total

Q1 · Q2 · Q3 · Q4 — due date · amount · paid · remaining

Record payment "Estimates for planning, not tax advice"

6 · Reports / Admin portal

Reports: Year ▾ Quarter ▾ → key totals → **Download tax summary PDF** · CSV

Admin: charts (sign-ups, invoices, expenses per week)

Admin: users table — role ▾ · active ☑ · audit log

6. Challenges and solutions

Challenge	Solution
Where to store receipt photos for free	Save them inside PostgreSQL (as bytes) — no paid file storage needed
"Overdue" could become out of date if saved	Work it out from the due date each time an invoice is read
Keeping screens simple on phones	Each wireframe has a single-column mobile layout; tables become cards

7. Plan for next week (Week 5)

- Set up the Git repository with branch protection and the project folders.
- Connect to the PostgreSQL database and run the first migration.
- Build secure sign-up and login (bcrypt password hashing, JWT session cookie).